

Original article

Techno-economic analysis of a hybrid system to power a mine in an off-grid area in Ghana

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ABSTRACT

In spite of the abundance of renewable energy and its potential application to mining industries, Ghana has not seen much investment in this area. The provision of electricity for mining activities in Ghana is cost intensive, especially in remote areas that are not connected to utility-grids. In this study, a technical and economic analysis of a hybrid electric power system for an off-grid mine company was conducted. Three different hybrid systems, together with a “Generator only” system as base case were analysed using HOMER software, to select the optimum energy system. The selected system consists of 50 MW of Solar PV, 15 MW of fuel cell system, 600 batteries, 20.5 MW of converter and 20 MW of diesel generator. The system produces 152.99 GWh of electric energy annually. The PV array produces 44%, the fuel cell produces 40% and the diesel produces 14%. At the current cost of system components, and based on assumptions adopted in this study, the optimal system yields a cost of energy of US\$0.25/kWh and a least cost of energy of US\$0.22/kWh, at a diesel price of US\$0.80/litre and 30% reduction in both PV module and fuel cell system costs.

Introduction

There is much demand for electricity in Ghana due to its growing economy, increase in per capita consumption of electricity by the addition of new consumers to the national grid, and the increase in modernisation. Sustainable economic growth of any developing country mainly depends on the supply of electricity infrastructure. To fulfil this ever-growing demand of energy sustainably, there is a need to exploit the renewable energy (RE) potential of the country. The industrial sector, of which the mining sector is part, consumes the highest amount of electricity, as opposed to residential and non-residential sectors in Ghana [1]. For instance, in 2014, the industrial sector consumed 5055 GWh (which represent 49.6%) out of 10 182 GWh consumed by all sectors, which are residential, non-residential, industrial and street lighting [2].

Mining is an energy-demanding industry, and energy is a critical concern for its effective operation and profitability. Analysis reveals that in 2009, the major gold mines consumed about 13% of Ghana's electricity; this translated into about 317,000 tCO₂ emissions (using a 2006–2008 weighted average grid emissions factor of 265 tCO₂/GWh) [3]. It was also estimated that in 2008, the major gold mining companies used about 9% of the country's diesel fuel consumption that year [3]. If little or non-carbon electricity sources are to replace the current

electricity sources, emissions would be reduced significantly.

Mining sites are, usually, sited in locations with substantial amount of ore deposits. These locations are mostly remote and isolated from suitable energy infrastructure. Mining companies often resort to off-grid energy systems for operation. Remote mines operating independently from the electrical grid usually rely on diesel generation as a preferred energy systems option because of their relatively cheaper upfront investment cost, notwithstanding the negative implications of using these energy systems. Examples of negative impact of these systems are noise pollution, emission of pollutants, and high maintenance and operating costs (mainly from increasing cost of fuel).

Therefore, mining industries are continually finding interest in renewable energy technologies (RETs) because most often than not, some RETs are more cost competitive than traditional energy solutions, especially in isolated off-grid locations, where miners often have to transport fuel over very long distances. The more isolated the mine, the more likely alternative energy sources are required [4]. A study conducted by CRMPs and THEnergy reveals that typical cost savings from solar photovoltaic range from 25% to 30%. In highly remote areas with escalated fuel prices, the reductions can go beyond 70% [5].

EYGM Limited [4] explained that renewable energy (RE) sources could help mining companies achieve energy security, minimise exposure to energy price fluctuations and increase energy price

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prediction. The role renewable energy resources will play in mining is not as far-off or futuristic as one may suppose. The debate for big investments in RE exceeds sustainability and social responsibility, and it is now a strong economic reason for mining companies [4]. With cost and operational efficiencies forming the backbone of mining businesses, innovative hybrid system solutions can provide RE as an alternative source of fuel, reducing fuel consumption, whilst providing reliable and affordable electricity. These systems could hypothetically save over three (3) million litres of fuel yearly [6,7].

The major drawback of these clean sources of energy is that, the initial investment can be extremely high. However, the costs of RETs like solar PV have been dwindling over the years due to technology advancement [8]. There is an increasing demand for these energy technologies. These technological innovations have led to annual declines in RE costs, and companies can take control of their energy requirements using on-site RE generation, efficiency technologies and micro-grids. This creates significant prospects for cost savings and operational optimisation [3].

Hybrid systems combine one RE source and a conventional source or more renewable forms of power with or without a conventional energy source to provide a particular energy service or energy requirement that works as a standalone system or tied to the grid. These systems may or may not include energy storage and backup components [9]. Several studies have been conducted on the application of hybrid energy systems around the world in such sectors as homes, small and large hotels, school buildings, remote villages, and other commercial buildings [10,11]. However, per the authors' knowledge, very little has been done on hybrid energy systems in Ghana and can be found in the open literature [12], and particularly, for the mining sector.

Notable efforts to diversify energy sources, and to increase the utilisation of RE resources have been increasing around the world [13]. Literature deliberate on different combinations of RE systems together with conventional diesel generation, with or without storage, and the evaluation of their technical and financial viability [13–15]. These literatures have found out that the combination of RETs with diesel generation result in lower cost of energy, increased reliability and reduced emissions of greenhouse gases (GHG), as well as lower cost of energy production as compared to systems consisting of only one energy source [16].

Adaramola et al. [12] presented an economic analysis of the viability of a PV/wind/diesel hybrid system in an isolated location of southern Ghana. An optimal system of 80 kW PV array, 100 kW wind turbine, 100 kW diesel generator, a 60 kW converter and 60 units of battery produced the mix of electricity to meet the load. It has been observed that the PV/wind/genset hybrid was the most economically feasible solution based on the net present cost (NPC) and levelised cost of energy (LCOE).

Mohammed et al. [17] reported the design of an optimal standalone hybrid PV/Fuel cell (FC) power system, without battery storage, to supply the electric load demand of the city of Brest, France. The HOMER-based optimisation study, using the total NPC clearly shown that the proposed hybrid power system, and, in particular, the FCs are a viable alternative to diesel generators as a non-polluting reliable energy source with a reduced total cost of maintenance. It has also been shown that a FC generator could efficiently complement a fluctuating renewable source such as solar energy to satisfy growing loads [17].

Balachander and Vijayakumar [18] also demonstrated with their cost-benefit analysis of FC–Solar PV hybrid system application for smart grid distributed generation system that hydrogen storage fuel system is economically less compared with battery storage system. This study further explored the significance of minimising the excess energy from the hybrid system to cut down the LCOE for hybrid RE system. The results revealed that the minimum cost combination leads to 89% of total electricity produced by PV and remaining 11% by FC.

Three different hybrid power system configurations and diesel

generator system for producing electric power in rural areas of Bangladesh have also been analysed [19]. The foremost aim was to determine the optimal size of systems that could meet the daily load demand of 50 households for three remote sites. The results showed that a hybrid system consisting of 6 kW PV, 10 kW diesel generating set, with battery storage was the most economically feasible, and by using the system, CO₂ emissions could be decreased by 38% over the diesel only system.

Salehin et al. [20] also designed an optimum hybrid system made up of solar PV and diesel generator. The purpose was to discover the optimum hybrid system based on available resources for a location in Bangladesh. The evaluation of electricity production, techno-economic analysis and CO₂ emissions analysis for the system was undertaken for the selected location. When the PV/diesel generator/battery system was compared with a system of diesel generator only, it was found out that addition of PV with diesel generators was appropriate and had a minor effect on the environmental.

The prospect of utilising hybrid standalone energy system for electricity generation in rural and semi-urban areas in the Northern part of Nigeria using HOMER software has been reported by Adaramola et al. [21]. The study suggests that a PV/Generator/Battery hybrid system is the most economically viable option for the selected locations, and other similar locations. Additionally, the simulation results indicate that the LCOE for the optimum hybrid system was below the cost of using diesel generator only. Reduced GHG emissions were also reported.

Many areas in Ghana have no access to electricity from the utility grid. There is a challenge to supply electricity because of two reasons. First, there is not enough power generation to fulfil the current power demand [1]. More so, even if there is enough power generation, extension of the grid to off-grid areas may be very challenging due to their geographical locations, and may not prove economical. ARE [22] reported that due to the usually, long distances between the existing grid and the off-grid areas, the costs of electrifying these areas through grid extension are very high, and, therefore, hybrid RE systems, in most cases offer a cheaper and less polluting option. Furthermore, the increased reliability of these systems and the best use of local resources favours this solution. Hence, hybrid RE systems prove to be the possible means of providing electricity for off-grid sites.

Regardless of the scale of operation, a mine could benefit from RE as a strategy to gain a measure of energy independence, not only from an unreliable grid, but also from the carbon-based fuels traditionally used on site. There are many renewable power projects that have proved successful in meeting the electrical load of a factory [6].

However, there is limited cases in Ghana, particularly, cases from the mining sector to demonstrate the technical feasibility and potential economic gains that could be realised from the application of a hybrid system in that context. Establishing the technical and economic feasibility of developing a hybrid system to provide electric power for mining activities will inform relevant stakeholders about options to reduce the load demand on Ghana's national grid, and make the excess power available to those with less capability to generate their own electricity. From an environmental perspective, options to reduce emissions, noise pollution and the mine's overall carbon footprint. This will also set the pace for projects of similar nature to be developed to serve the mining and other industries. Consequently, the overall objective of this study is to perform a technical and economic feasibility study employing hybrid RE system to power an off grid mine in Ghana.

To achieve the overall general objective of this study, the following specific objectives were also formulated: (i) determine the load profile of a selected mine and size an appropriate power plant; (ii) determine which option is both technically and economically feasible, if any; and (iii) analyse the sensitivity of the hybrid power system. The scope of this work is limited to the development of a framework to aid the planning, technical and economic analysis of an off-grid hybrid renewable energy system. The work, however, does not include the

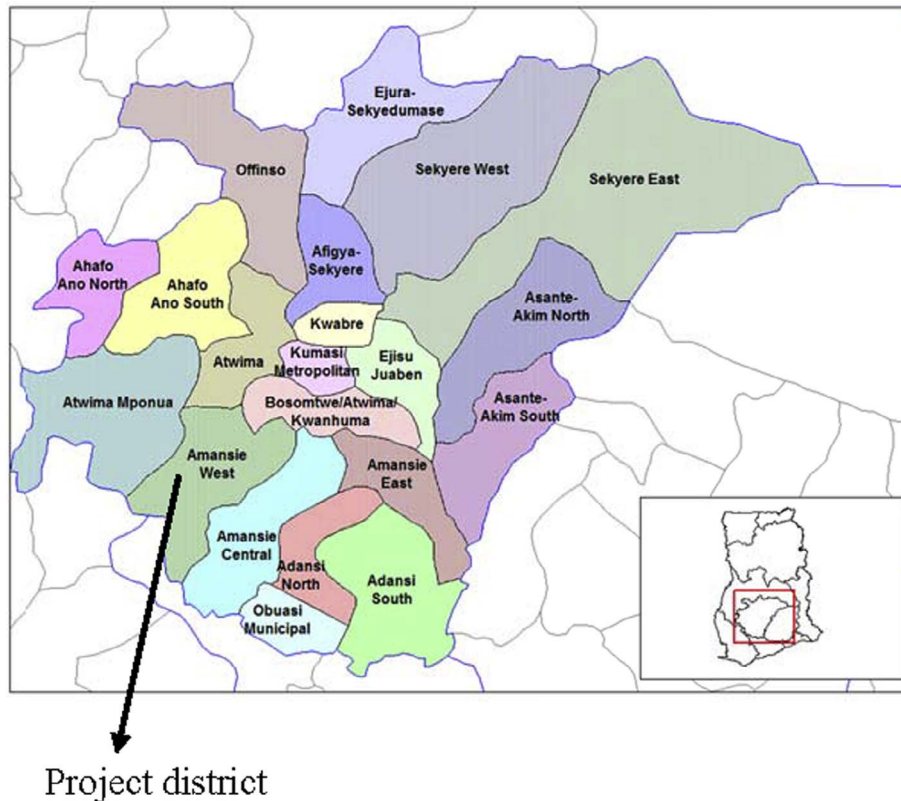


Fig. 1. Map of Ghana showing Ashanti region's districts.

detailed design of the hybrid renewable energy system. The combination of RETs together with fossil fuel generators and storage devices as optional backup is possible, though some of them may not be most efficient, cost effective and reliable.

Materials and methods

The design of a stand-alone hybrid system is site specific and depends on both the resources available and the load demand. The approach to decide the most suitable hybrid system involves gathering field data for the site and determining which components to use [23].

Site location and electrical load

The site considered for this study is an off-grid site in Ghana, with gold deposits. The site is located on latitude $6^{\circ} 30' N$ and longitude $2^{\circ} 4' W$, in the Amansie West district of the Ashanti region of Ghana (see Fig. 1), approximately about 80 km southwest of Kumasi. The site is within a region that receives a daily solar irradiation ranging from 4.5 to 5.3 kWh/m², and has a monthly average wind velocity of 3.2–3.8 m/s² at a height of 50 m.

This site was selected because there is a new gold mine project currently underway around the site. The mine is isolated with no access to the grid. Its current main source of electricity is diesel generator sets. The 20 MW power package, consisting of 20 units of 1000 kW generator sets is to supply power to the mine until a permanent grid power is established.

The average power demand of the off-grid mine was estimated in consultation with engineers and data gathered during site visits. A continuously running mine with an electric load of about 15 MW is estimated, based on a typical mine with a connected load of 20 MW, with peaks of about 19 MW. This load has been scaled by Hybrid Optimisation Model for Electric Renewable (HOMER), a software used in this study, to an annual value of 350 MWh/day, with a scaled peak load of about 19 MW. Fig. 2 shows the daily load profile of the off-grid

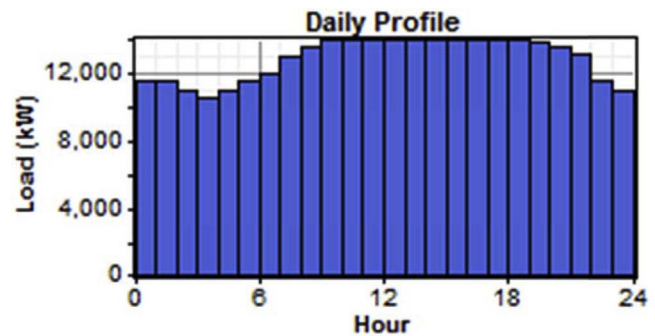


Fig. 2. Daily electricity load profile for the off-grid mine.

mine.

Modelling and simulation in HOMER

The study proposes to use HOMER to model and simulate the hybrid system. The proposed hybrid system as modelled in HOMER for simulation to select the most technically and economically feasible energy system is shown in Fig. 3.

HOMER was originally developed at the National Renewable Energy Laboratory (NREL). It was later enhanced, and is distributed by HOMER Energy. The software uses numerical calculation for simulation, and is more effective for micro power systems. The program is able to simulate a vast number of scenarios, including mixing conventional generators with RETs to meet demand. Scenarios can be analysed as either grid tied or standalone, and their emissions levels estimated. HOMER allows the user to make inputs of RE generation to the required demand. This allows the user to analyse potential small-scale power systems before implementation. It also allows comparing many different design options with regards to their technical and economic benefits [24].

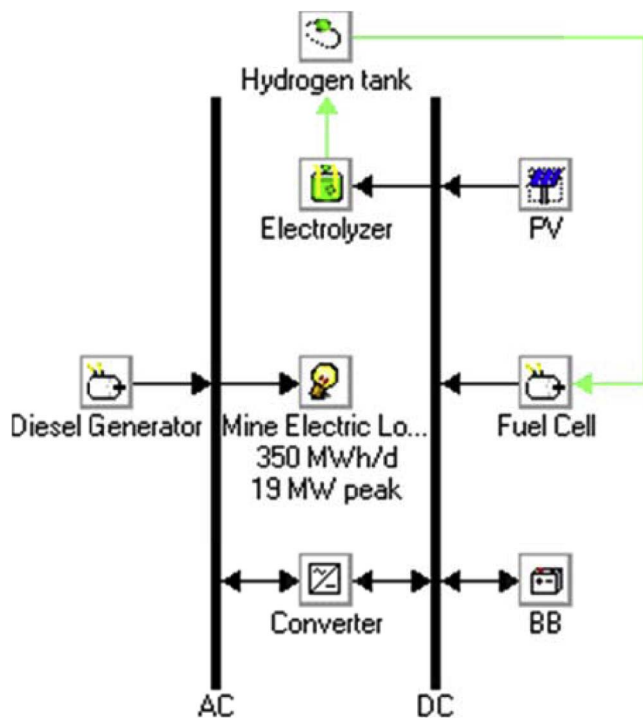


Fig. 3. HOMER Model of the Hybrid Power System.

Resources input

HOMER requires input data in order to model, simulate and analyse a system's performance. Sections "Resources input" to "Other inputs" discusses the various inputs. The resources defined for HOMER in this study are the solar resource and diesel fuel. The solar resource data of the site located on latitude 6°30'N and longitude 2°4'W in the Amansie West district of the Ashanti region of Ghana and essential information on diesel fuel in Ghana are presented in subsequent sections

Solar radiation data. Ghana is endowed with vast renewable energy resources, particularly biomass, solar, and wind energy resources, spread across the entire nation. Sites along the coast and high elevations show moderate to excellent wind energy potentials, especially, along the east coastlines, although wind energy may not be adequately exploited. However, the sites with moderate wind resource for power generation lies along the coast of Ghana. Wind energy has not been considered for the hybrid system of this work. Ghana receives annual daily averaged global solar irradiation ranging from 3.1 kWh/m² to 6.5 kWh/m² across the entire country. The annual sunshine duration ranges between 1800 and 3000 hours, giving a very high potential for grid connected and off-grid solar systems [25].

Solar radiation data are essential elements in PV output calculations. The amount of solar radiation on the surface of a solar device depends on the positioning and inclination of the device [25]. For every location, there is an optimal orientation and inclination angles, which results in maximum amount of solar radiation received by a solar collector. For locations in the northern hemisphere like Ghana, it is paramount to position the solar device to face south at an inclination angle of 10–30° [26].

The solar radiation data used in this study is satellite data obtained from NASA Surface Meteorology [27]. This choice has been made because there was no publicly available ground measured solar radiation data for the selected site. Fig. 4 shows the monthly daily averaged global solar radiation for this site. Monthly variation in global solar radiation is observed, and hence, the monthly energy output from solar energy conversion systems would vary from one month to another. It can also be observed from Fig. 4 that the minimum daily global solar

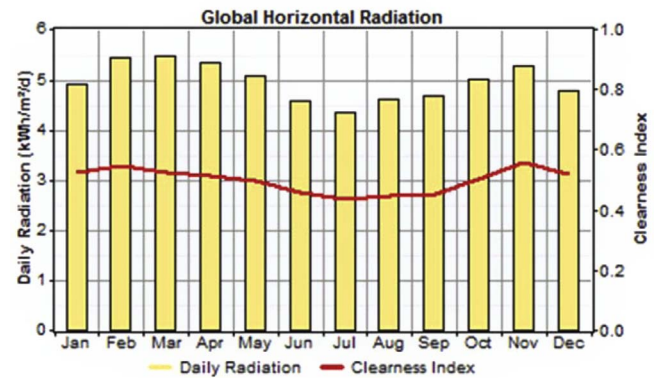


Fig. 4. Horizontal Global Solar Radiation.

radiation was 4.34 kWh/m²/day (in July) while the maximum value was 5.45 kWh/m²/day (in March); with annual averaged daily global solar radiation of 4.95 kWh/m²/day.

Diesel fuel. Diesel fuel inputs include cost, density, carbon content and lower heating value (LHV). Historical fuel cost data was obtained from the National Petroleum Authority (NPA) of Ghana [28]. The price of diesel fuel in Ghana was US\$0.80/litre when this study was initiated (November 2015). However, in remote areas, diesel prices may be significantly higher than that of the official price, due to transportation and other ancillary costs. In order to investigate the effect of changes in fuel price on the optimal hybrid energy system and cost of electricity produced by the power system, diesel price was, therefore, varied between US\$0.8/litre and US\$1.5/litre. The highest price of US\$1.50/litre, which is almost twice the official price, caters for the possible extreme price of diesel in remote areas or a sudden spike in the price. The fuel density, carbon content and lower heating value for the diesel fuel are 820 kg/m³, 88% and 43.2 MJ/kg, respectively.

Specification and cost of components

To facilitate the modelling and simulation of the system components, HOMER requires cost data and specification parameters of each system's components. To obtain the input data utilised by HOMER in modelling the various hybrid systems, data were collected from journals, written reports, websites, government agencies, various service providers, and personal communication with representatives and engineers from dealers and equipment manufacturers [29–33].

Photovoltaic panels. The typical average cost of a complete PV system from recent installations in Ghana is about US\$4035/kW based on communication with local PV system dealers [30,31]. The PV modules account for around 55% of this cost. Therefore, the initial and replacement cost of the PV panels, including installation is taken as US\$2200/kW. These prices are likely to vary from the actual system quotes due to many market factors (which are, dealer's profit mark-ups, size of the solar panel, technology, the brand etc.), and are therefore only indicative. The operating and maintenance cost is taken as US\$10/year [34] (mainly for cleaning the PV system) which may not be applicable due to occasional rains. The panels are modelled as fixed and tilted toward south at an angle of 15° to avoid settlement of dirt. The lifetime of the PV array (which is the same as the project life of the proposed hybrid energy systems) is taken to be 25 years [17,21,35]. The derating factor, which compensates for efficiency reduction in the PV performance due to temperature, dust, and wiring losses is taken as 0.75 [17,36] and the ground reflectance is taken as 20% [26] in this study. For simulation analysis, three different PV array sizes, which were 0 (no PV array), 50 MW and 60 MW are considered. The high installed capacity is used in order to produce excess electricity to produce hydrogen fuel to run fuel cell generator as well as charge the

battery bank, to be used during the night and periods of poor weather conditions.

Fuel cell system. The fuel cell system is modelled in HOMER as a generator, which generates DC electricity using stored hydrogen as fuel. The hydrogen is produced from an electrolyser using power from the PV system. The initial capital cost of 1 kW unit of the generator was taken US\$1200/kW [32]. The replacement cost was assumed to be the same as the initial capital cost for the generators, while the operational and maintenance cost is assumed to be US\$0.050/h/kW for possible replacement of membrane during the life of the FC [37]. For simulation analysis, 0, 10 MW and 15 MW sizes of fuel cell DC generator are considered. This allows the fuel cell to meet the base load in the absence of other energy sources at a reasonable cost since larger sizes would increase the overall system cost. Kalinci et al. [35] considered a life time interval of FC generator as 30,000 h, 40,000 h and 50,000 h and realised a marginal decrease in the LCOE respectively. Therefore, the operating lifetime of the FC generator was taken as 50,000 h for this study. The fuel curve intercept and slope of the fuel consumption curve are 0.0 litre/hr/kW and 0.06 litre/hr/kW, respectively.

The maintenance cost was assumed to be US\$100/year [17]. For simulation analysis, 0, 34 MW and 36 MW sizes of DC electrolyser are considered to produce enough hydrogen for storage to meet the demand of the fuel cell. The efficiency of the electrolyser is assumed to be 85% and the lifetime of an electrolyser is taken as 15 years [17]. The hydrogen produced by the electrolyser is stored in hydrogen tank. Four different sizes of hydrogen tank are used for simulation: 0 tonne, 3000 tonnes, 32,000 tonnes and 33,000 tonnes. The investment and replacement costs of 2 tonnes of a hydrogen tank is taken as US\$1000 while the operational and maintenance cost is assumed to be US\$5/year [17]. The lifetime of each tank is assumed to be 25 years.

Batteries. The input parameters of batteries are the cost and the number of batteries required. The following are also defined for the batteries: nominal capacity, nominal voltage, round trip efficiency, minimum state of charge, float life, maximum charge rate, maximum current and life-time throughput. In this study, the Hoppecke vented lead-acid with a nominal capacity of 3000 Ah, cell voltage of 2 V, lifetime throughput of 10,231 kWh was used because these batteries are an inexpensive option and have low maintenance requirement. In a real installation, a smaller number of larger batteries may be preferable, and economic analysis performed by HOMER would not be significantly affected by this distinction [38]. The capital and replacement cost of one battery is taken as US\$350 [30,33] and the operational and maintenance cost is taken as US\$5/year [30]. For simulation analysis, two different sets of quantities of batteries at a voltage of 48 V (12 batteries per string), which are 0 (no battery), 10 strings and 25 strings, were considered just to store any excess energy from PV after the electrolyser have been served.

Converter. A power converter maintains the flow of energy between the AC electrical load and DC components of the hybrid energy system. The capital and the replacement costs of 1 kW converter were taken as US \$850, which is about 20% of a complete PV system installation from recent installations based on communication with local PV system dealers. Similarly to the cost of PV, the price of the converter is indicative are likely to vary from the actual system quotes due to many market factors and are therefore only indicative. In addition, the operational and maintenance cost is taken as US\$100/year [17]. For simulation analysis, four different sizes of converter (0 MW, 10.5 MW, 20.5 MW and 30.5 MW) were considered based on the PV and FC sizes. The inverter and rectifier efficiencies are assumed to be 90% and 85% respectively for all sizes considered as suggested Givler and Lilienthal [38]. The lifetime of a unit is taken as 15 years since selected manufacturer gives a 10-year warranty [30,39].

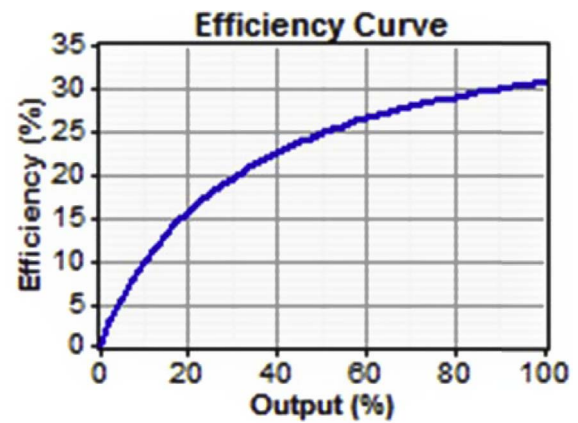


Fig. 5. The fuel efficiency graphs for selected generator.

Diesel generator. The diesel generators are used to supplement the power production by the PV energy conversion system especially when the required electrical load is not met by it, the battery bank or fuel cell system. The cost, operation and maintenance costs used are from local dealers in generators in Ghana [31,32]. The initial capital cost for 1000 kW unit of a diesel generator is US\$265000. The replacement cost was assumed to be the same as the initial capital cost for the generators, while the operational and maintenance cost was assumed to be US\$0.50/h. For simulation analysis, four different set of 1000 kW generator (0, 10 units, 20 units and 25 units) are used in other to provide a back-up for the base load in the absence of the RE sources. The operating lifetime of each generator was taken as 40,000 h. The efficiency curve and the fuel curve for the diesel generator are shown in Fig. 5 and Fig. 6 respectively. The minimum load ratio was taken as 30% [38]. Loading the generator below 30% of rated power could lead to high energy loss. The fuel curve intercept and slope of the fuel consumption curve are 0.08 litre/hr/kW and 0.25 litre/hr/kW.

Other inputs

In addition to resource and system's component inputs, HOMER further requires economic, constraints, emission and system control input parameters. To determine the NPC of the system, economic parameters like annual real interest rate, project lifetime, system fixed capital cost, system fixed operating and maintenance (O & M) cost, and capacity shortage penalty must be specified. The lifetime of the proposed system is assumed to be 25 years, which is typical; the annual real interest rate is taken to be 7.5%. The real interest rate is equal to the nominal interest rate minus the inflation rate [38]. The current nominal interest rate and inflation rate as of 2015 were 26% and 18.5% respectively [32,40]. Adaramola et al. [12] estimated US\$50,000 as initial fixed capital cost which can be used for site preparation and other ancillary activities. For the current system, US\$100,000 was assumed as the system fixed capital cost due to the relative size of the

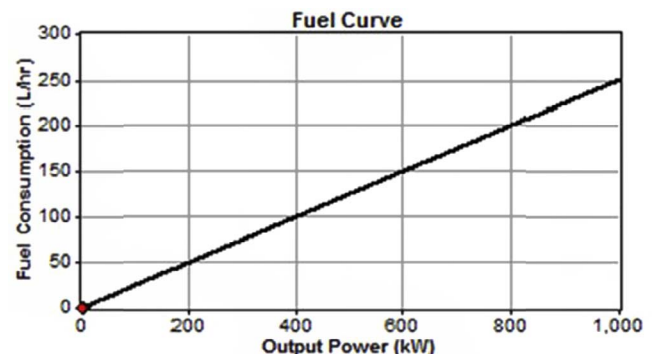


Fig. 6. The fuel consumption curve for selected generator.

project.

Constraints are conditions, which the systems must satisfy. HOMER discards systems that do not satisfy the specified constraints, so they do not appear in the optimization results or sensitivity results [24]. For this study, the constraint inputs considered were: the operating reserve taken as percentage of hourly load of 10% as suggested by Cotrell [37] and Dalton et al. [41]. The maximum annual capacity shortage and minimum renewable fraction (RF) were taken as 0% respectively to allow only the diesel generators to operate when RE power is unavailable. The operating reserve taken as percent of renewable output for solar power output was 25% [24].

Cost penalty associated with a pollutant, or a limit on the emissions of a pollutant can be specified for a system. HOMER allows emission penalties and emission limits on carbon dioxide, carbon monoxide, unburned hydrocarbons, particulate matter, sulphur dioxide and nitrogen oxides [24]. Since there were no emissions penalties in Ghana at the time of the study, emission inputs were all set to zero.

System control inputs define how HOMER models the operation of the battery and generators. Multiple generators have been allowed to run simultaneously. HOMER can model two dispatch strategies, cycle charging and load following. Under the load-following strategy, the generator provides only the power necessary to meet the load at the time. With the cycle-charging strategy, once the generator is operating, it produces as much power as possible to charge the batteries in addition to meeting the load [24]. The load following strategy was chosen for simulation.

Sensitivity analysis

Sensitivity analysis of the system is required under different levels of changes for the vital input parameters such as capital costs of components, fuel cost, etc. Sensitivity analysis of a system is achieved by systematically altering the values of a particular parameter under study. The results of the various parameters are afterwards assessed, and their effects on the viability of the system compared. The sensitivity of the input variables chosen informed how the system behaved to changes in these variables. This allowed an understanding of the effects of uncertainty, and make good design decisions even under conditions of uncertainty. For the sensitivity analysis, the price of diesel fuel, cost of the PV system and fuel cell system were varied as shown in Fig. 7.

Results and discussion

Optimal hybrid energy systems (Scenarios)

The results of the simulations show the feasible combinations of system components that satisfy the electric load inputs and system constraints. Four scenarios were chosen and compared to select techno-economic system based on the least NPC, LCOE, renewable fraction (RF) and diesel fuel consumption and hence, minimum pollution to the environment. These scenarios are:

- Scenario – A: PV, FC, Diesel Gen-set and Battery;
- Scenario – B: PV, FC and Diesel Gen-set;
- Scenario – C: PV and Diesel Gen-set;

Table 1
Component sizes of scenarios considered.

	Scenario A	Scenario B	Scenario C	Scenario D
PV Array	50 MW	50 MW	50 MW	none
Fuel Cell	15 MW	15 MW	none	none
Electrolyser	30 MW	30 MW	none	none
H2 Tank	25,000 tonnes	15,000 tonnes	none	none
Converter	20.5 MW	20.5 MW	20.5 MW	none
Batteries	600	none	none	none
Generator Sets	20 MW	20 MW	20 MW	20 MW

Table 2
Comparison of results.

Scenario	NPC (Million US\$)	LCOE (US \$/kWh)	RE Fraction (%)	Diesel Consumption (Million L)	Emission (kt CO ₂ /yr)
A	345.10	0.242	85	8.20	21.56
B	402.68	0.283	69	17.09	44.98
C	531.97	0.374	38	39.93	105.95
D	426.67	0.300	0	45.95	121.01

- Scenario – D: Diesel Gen-set only.

The scenarios A, B and C are possible alternatives to scenario D, which correspond to the current source power generation at the selected mining site. The optimum sizes of the chosen scenarios are presented in Table 1, and a summary of various comparison metrics of these scenarios are listed in Table 2.

The diesel only system (Scenario D) had the least initial capital cost, but ended up with a greater NPC compared to scenario A. The hybrid system configurations (Scenarios A to C) considered required a huge initial capital, but in the end, they gave a less NPC because of lower fuel consumption. It is interesting to note, that Scenario A shows the lowest LCOE (of US\$0.24/kWh), and the highest the portion of power generated by RE sources (85%). The second lowest LCOE of US\$0.28/kWh is for scenario B. The reason for the increase in LCOE for the other systems (Scenarios B and C), with respect to Scenario A, is due to reduction in the contribution of RE sources and increasing consumption of diesel fuel. The highest LCOE and the least RF belong to Scenario C and Scenario D, respectively. The declining in the cost of RETs and the flexibility of using of hydrogen as the stored to meet daily load, has increased the viability of the Scenario A and decreased dependence diesel generation.

Selected hybrid energy system

After comparing the scenarios, the configuration (Scenario A) with the least NPC, least operating cost, least LCOE, high renewable energy fraction and less diesel fuel consumption and minimum emission of pollutants into the atmosphere was considered as the most appropriate option from the technical and economic point of view.

The monthly average electricity production by the different energy generators of the selected configuration (Scenario A) are shown in

Fig. 7. Sensitivity analysis values.

	Diesel ([\$/L])	PV Cap. (Mult.)	FC Cap. (Mult.)	FC Repl. (Mult.)
1	0.800	1.00	1.00	1.00
2	1.000	0.90	0.90	0.90
3	1.200	0.80	0.80	0.80
4	1.500	0.70	0.70	0.70

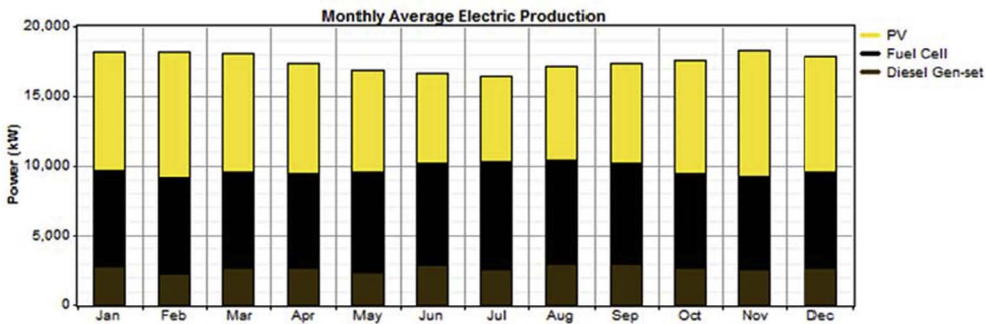


Fig. 8. Monthly average electric energy production.

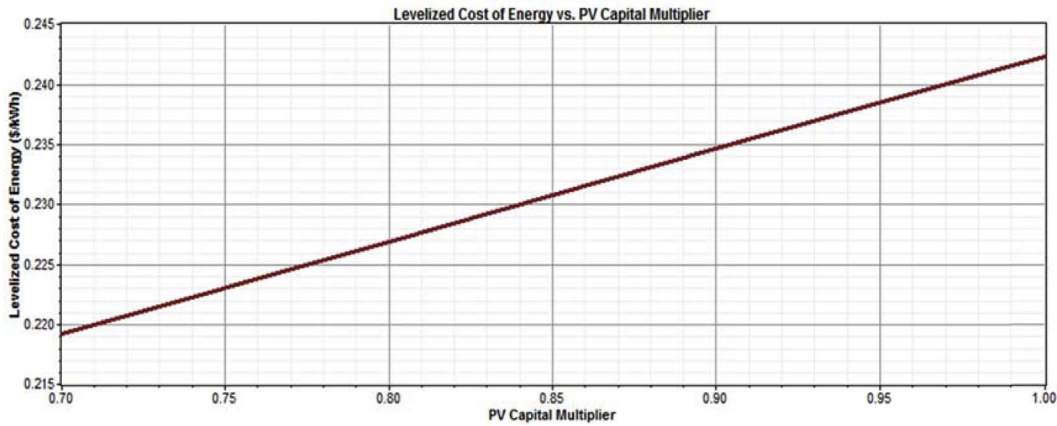


Fig. 9. Sensitivity analysis for PV capital cost at a fixed cost of diesel and Fuel cell.

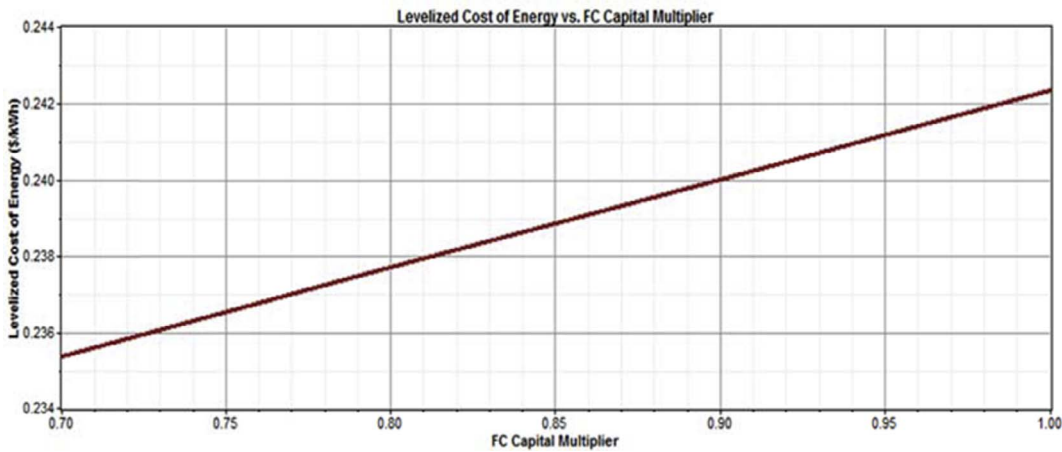


Fig. 10. Sensitivity analysis for fuel cell capital cost at a fixed cost of diesel and PV.

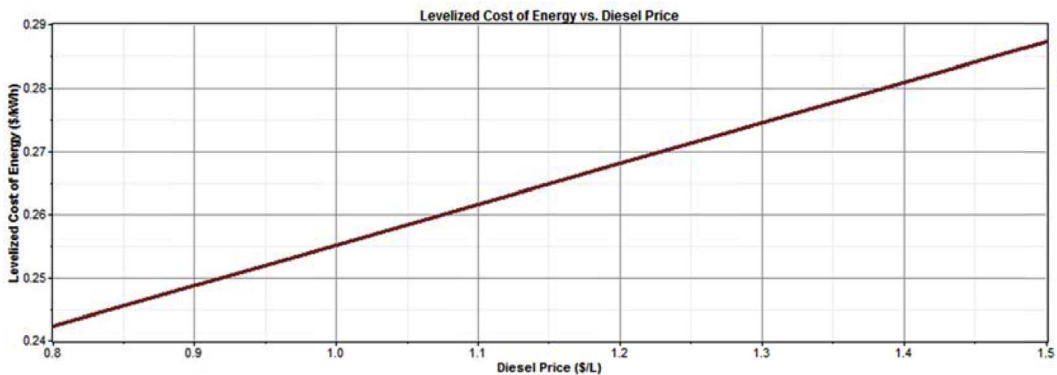


Fig. 11. Sensitivity analysis for cost of diesel at a fixed cost of PV and FC systems.

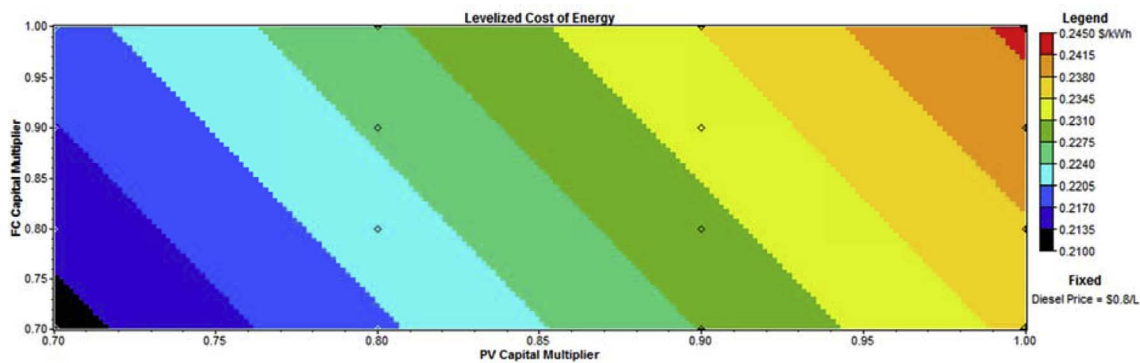


Fig. 12. Sensitivity of analysis of both PV and FC capital cost at a fixed cost of diesel.

Fig. 8. The PV array produces 44% of the annual total electricity of 152.99 GWh/year followed by the fuel cell, which produces 40%, and 15% of the annual electricity is produced by the diesel generator. Eighty-seven percent of the total energy produced by the PV serves the AC primary load and the electrolyser uses remainder to produce hydrogen.

Sensitivity analysis

Sensitivity analysis has been performed on PV and fuel cell capital cost input, and diesel fuel price for the selected scenario (scenario A). The result of reducing the PV system cost by 10%, 20% and 30% is presented in Fig. 9. This reduction could be possible due to many factors, which may include technology enhancement, increasing demand, capital subsidy, tax credit and incentive. Reducing the PV system cost makes the project more viable by producing electricity at lower LCOE with respect to system with base module PV cost (with price reduction). The LCOE are US\$0.235/kWh, US\$0.227/kWh and US\$0.219/kWh respectively for 10%, 20% and 30% in reduction in module cost. These values are comparable to the current approved electrify tariffs for mines in Ghana. At the time of the study, mining industries were being charged the Ghana Cedis equivalent of US\$ 0.296/kWh (exclusive of max demand charge of US\$1.54/kVA/month, monthly service charge US\$1.54 and 17.5% VAT) for electricity consumed from the grid [42].

Fuel cell capital cost is also likely to decline resulting from technology advancement, high demand and mass production [43]. Fig. 10 shows that the LCOE is reduced to US\$0.240/kWh, US\$0.238/kWh and US\$0.235/kWh for the 10%, 20% and 30% decrease in fuel cell capital cost.

Fig. 11 shows the effect of fuel price on the LCOE of the hybrid system. The LCOE increases from US\$0.248/kWh to US\$0.287/kWh for a diesel price of US\$1.50/litre. It should be mentioned that the LCOE of Scenario D (diesel generator only) at diesel price of US\$ 1.50/litre is about US\$0.551/kWh. This makes Scenario A more economically profitable than the conventional system (Scenario D), which is usually employed in off-grid mine sites.

Fig. 12 shows different LCOE regimes for situation whereby there is reduction in both the capital cost of PV and fuel cell for Scenario A. It can be observed for example, that LCOE for the Scenario A is US\$0.212/kWh at a fixed diesel price of US\$0.80/litre and 30% reduction in capital cost of both PV and fuel cell systems. As the capital cost of both PV and FC reduces at a fixed diesel price of US\$0.8/litre, the initial capital and NPC reduces correspondingly and hence, reduced LCOE.

Conclusions

A technical and economic assessment of a hybrid electric power supply system for an off-grid mine has been conducted. Solar irradiation data for the site has been obtained from NASA Surface Meteorology and HOMER software was used to find the optimal size of the system

components by comparing the total NPC and LCOE for different configurations. The results show that solar energy potential in Ghana could be exploited for the provision of electricity in mine industry.

Three different renewable-based hybrid systems have been considered together with a “Generator only” system as base case. The hybrid systems comprise any of the following energy generators with or without storage: solar PV, fuel cell system and a diesel generator sets. The final selection of the techno-economic system is based on the following criteria: minimum total NPC and operating cost, lowest LCOE, high renewable fraction, and minimum diesel fuel consumption with a corresponding minimum emission of pollutants into the atmosphere. The configuration (Scenario A) with 50 MW Solar PV, 15 MW of fuel cell system (which comprises 34 MW electrolyser and 25,000 tonnes of hydrogen storage tank), 600 batteries, 20.5 MW converter and 20 MW of diesel generator sets is found to be the best option. The technical and economic analysis show that this option produces 152.99 GWh of electric energy per year. The PV array produces 44% of this energy, the fuel cell produces 40% and the diesel produces 14%. Out of this production, the mine load consumes 127.75GWh per annum, which constitute 87%, and the remainder serves the electrolyser for the production of hydrogen. At the current cost of system components (at the time of this study) and constraints, the system yields the LCOE of US\$0.212/kWh at a diesel price of US\$ 0.80/litre and PV array investment cost of US\$1540/kW and fuel cell system cost of US\$840/kW. This LCOE is over 28% lower than the current approved tariff for mines in Ghana [42].

In spite of the abundance of RE and its possible application to mining industries, Ghana has not seen much investment in this area due high initial cost associated with such technologies. It is expected that the framework developed in this study will make it easy for stakeholders in the mining and allied industries to analyse and consider hybrid systems for application in the mining sector because of the declining of cost of renewable energy systems, and environmental pollution concerns.

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References

- [1] Kemausor F, Mensah GS, Addo A. Ghana energy access review (GEAR) report. Kumasi, Ghana: Kwame Nkrumah University of Science and Technology; 2014.
- [2] Energy Commission of Ghana. National Energy Statistics 2005 – 2014 2015:15. Available on: http://energycom.gov.gh/files/EnergyStatistics_2015.pdf [accessed May 15, 2017].
- [3] Würtenberger L, Hassan E. Policy brief: low carbon options in the gold mining

- industry in Ghana 2011:4. <http://www.ecn.nl/docs/library/report/2011/o11023.pdf%0A> [accessed October 5, 2015].
- [4] EYGM Limited. Renewables in mining: futuristic or realistic? 2014:8.
 - [5] Cronimet/THEnergy. Cronimet/THEnergy study: in solar for mines size does not always matter 2015:1–2.
 - [6] Eccles J. Don't let the sun go down. *Inside Min* 2014;7:50–1.
 - [7] Reddy V, Jorgensen P. African mining goes green. *Inside Min* 2014;7:54–5.
 - [8] REN21. Renewables 2010 global status report. Paris, France; 2010.
 - [9] Ibrahim M, Khair A, Ansari S. A Review of hybrid renewable energy systems for electric power generation. *Int J Eng Res Appl* 2015;5:42–8.
 - [10] Santika WG, Sunu WP, Arsawan MI. Feasibility analysis of a grid-connected PV system for home application. The 3rd Bali International Seminar on Science and Technology (BISSTECH), Grand Inna Kuta, Bali, Indonesia; 2015, p. B5.5-1-B5.5-6.
 - [11] Santika WG, Sudirman, Suamir IN. Feasibility analysis of grid/wind /PV hybrid systems for industrial application. In: Proceeding of the 2nd international conference on mechanical engineering. Bali; 2015.
 - [12] Adaramola MS, Agelin-Chaab M, Paul SS. Analysis of hybrid energy systems for application in southern Ghana. *Energy Convers Manage* 2014;1–37.
 - [13] Lau KY, Yousof MFM, Arshad SNM, Anwar M, Yatim AHM. Performance analysis of hybrid photovoltaic/diesel energy system under Malaysian conditions. *Energy* 2010;35:3245–55. <http://dx.doi.org/10.1016/j.energy.2010.04.008>.
 - [14] Al-Badi AH, Bourdoucen H. Study and design of hybrid diesel – wind standalone system for remote area in Oman. *Int J Sustain Eng* 2011:1–10. <http://dx.doi.org/10.1080/1478646X.2011.554608>.
 - [15] Shaahid SM, El-Amin I, Rehman S, Al-Shehri A, Ahmad F, Bakashwain J. Dissemination of off-grid hybrid wind- diesel-battery power systems for electrification of isolated settlements of hot regions. *Int J Sustain Eng* 2007;26:91–105.
 - [16] Olatomiwa L, Mekhilef S, Huda ASN, Sanusi K. Techno-economic analysis of hybrid PV – diesel – battery and PV – wind – diesel – battery power systems for mobile BTS : the way forward for rural development. *Energy Sci Eng* 2015;3:271–85. <http://dx.doi.org/10.1002/ese3.71>.
 - [17] Mohammed OH, Amirat Y, Benbouzid M, Feld G, Tang T, Elbast A. Optimal design of a stand-alone hybrid PV/fuel cell power system for the city of Brest in France. *Int J Energy Convers* 2014;2:1–7.
 - [18] Balachander K, Vijayakumar P. Economic analysis, modeling and simulation of photovoltaic fuel cell hybrid renewable electric system for smart grid distributed generation system. *Int J Mech Eng Technol* 2012;3:179–86.
 - [19] Mondal AH, Denich M. Hybrid systems for decentralized power generation in Bangladesh. *Energy Sustainable Dev* 2010;14:48–55. <http://dx.doi.org/10.1016/j.esd.2010.01.001>.
 - [20] Salehin S, Rahman M, Islam AKMS. Techno-economic feasibility study of a solar PV-diesel system for applications in northern part of Bangladesh. *Int J Renewable Energy Res* 2015;5:1220–9.
 - [21] Adaramola MS, Paul SS, Oyewola OM. Assessment of decentralized hybrid pv solar-diesel power system for applications in northern part of Nigeria. *Energy Sustainable Dev Assess* 2014;2014(19):72–82.
 - [22] ARE. Hybrid power systems based on renewable energies: a suitable and cost-competitive solution for rural electrification; 2008. http://www.ruralelec.org/fileadmin/DATA/Documents/06.Publications/Position_papers/ARE-WG_Technological_Solutions_-_Brochure_Hybrid_Systems.pdf. [accessed September 27, 2011].
 - [23] Ani VA, Nzeako AN. The potential of stand-alone PV/wind hybrid energy system for power supply to remote rural areas in Nigeria. *Int J Clean Environ* 2012;13:1–20.
 - [24] Lambert T, Gilman P, Lilienthal P. Micropower system modeling with HOMER. In: Farret FA, Simoes MG, editors. *Integration of alternative sources of energy*. John Wiley & Sons, Inc.; 2006.
 - [25] Ministry of Energy. National Energy Policy. Accra: Ministry of Energy; 2009.
 - [26] Quaschnig V. *Understanding renewable energy systems*. London: EarthScan; 2005.
 - [27] NASA. Surface meteorology and solar energy 2015. eosweb.larc.nasa.gov/sse [accessed March 31, 2016].
 - [28] National Petroleum Authority. <http://www.npa.gov.gh/indicative-price-year/indicative-prices-2015>.
 - [29] Cummins Gh. Ltd. <http://africa.cummins.com/locations/ghana/accra-ghana-head-office>.
 - [30] Energiebau Sunenergy. <http://www.ggee.net/members/energy/energiebau-sunenergy-ghana-ltd-24.html>.
 - [31] Electrofax Engineering Services. www.electrofax.com.
 - [32] BALLARD. www.ballard.com.
 - [33] HOPPECKE Batterien GmbH & Co. KG. <https://www.hoppecke.com/en/n.d>.
 - [34] Keshtkaran E, Fardoun AA, Noura H. Employing hydrogen fuel cell in hybrid energy systems for stand-alone, off- grid remote sites in UAE. In: International conference on renewable energies and power quality (ICREPQ'14), 8th to 10th April, 2014, Cordoba, Spain: Renewable Energy and Power Journal; 2014, p. 7.
 - [35] Kalinci Y, Hepbasli A, Dincer I. Techno-economic analysis of a stand-alone hybrid renewable energy system with hydrogen production and storage options. *Int J Hydrogen Energy* 2014;1–13. <http://dx.doi.org/10.1016/j.ijhydene.2014.10.147>.
 - [36] Quansah DA, Adaramola MS, Appiah GK, Edwin IA. Performance analysis of different grid-connected solar photovoltaic (PV) system technologies with combined capacity of 20 kW located in humid tropical climate. *Int J Hydrogen Energy* 2016;42:4626–35.
 - [37] Cotrell J, Pratt W. Modeling the feasibility of using fuel cells and hydrogen internal combustion engines in remote renewable energy Systems. Technical Report; 2003.
 - [38] Givler T, Lilienthal P. Using HOMER ® software, NREL's micropower optimization model, to explore the role of gen-sets in small solar power systems case study : Sri Lanka Technical Report [NREL/TP-710-36774]; 2005.
 - [39] SMA Solar Technology AG. <http://www.sma.de/en.html>.
 - [40] Bank of Ghana. <https://www.bog.gov.gh/statistics/inflation>; 2016.
 - [41] Dalton GJ, Lockington DA, Baldock TE. Feasibility analysis of stand-alone renewable energy options for a large hotel. *Renew Energy* 2008;33:75–90.
 - [42] Public Utilities Regulatory Commission (PURC). Publication of Electricity Tariffs; Accra, Ghana: Ghana Gazette; 2015.
 - [43] 4th Energy Wave Ltd., The Fuel Cell and Hydrogen Annual Review, 2015; 2015.